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Semester/Exam Project

A Tiny Machine Learning Project

Purpose:

The purpose of this project is to apply certain theoretical topics, from the course, onto a specific project.

The project runs through the entire semester.

You must design an embedded artefact, composed of embedded hardware (the Photon2) embedded software, one or more sensor(s), suited preprocessing and a Machine Learning / Deep learning algorithm.

Your device shall be capable of predicting either :

1. A class outcome (classification)
2. A continious variable (regression)
3. Outliers / Anomaly detection

Maybe a combination of 1,2 and 3, anomaly detection.

You can seek inspiration, or ideas, for instance here:

<https://www.hackster.io/search?i=projects&q=tinyml> or <https://www.edgeimpulse.com/blog/category/projects/>

Your device could be anything from "*automated cat feeder, that recognizes your cat based on gestures or an image*" to a "*machine condition monitoring system, capable of recognizing certain vibration patters*" or something completely different.

What's essential in this project is that the task is not easily solvable, or impossible, using traditional rule based programming, and that you collect and label your own data.

Documentation:

A written project report must be handed in, including as minimum

- Introduction
- Project description
- Requirements analysis
- System design
- Implementation
- Test/Verification
- Conclusion

You should aim for app. 20 pages.

The project report must be uploaded to the exam-portal, in **pdf** format.

In the report you are welcome to include links to videos, that contain supporting information, such as demonstrating your artefact is working, selected verification tests executed or other relevant supporting information. The **pdf** report must contain the primary information.

Teamwork:

You are allowed to work in teams, but as we are a small class, consider to build you own project, so we get some diversity.

In your project documentation, is must be clearly stated what topics the individual team members have worked with.

Requirements:

Functional requirements

1. Your device must be able to read data from a connected sensor, local to the device
 1. A sensor can be anything that quantifies a physical measure, into an electrical signal, such as temperature, light, humidity, presence, movement, magnetism, pollution, etc.
2. Your device must be able to predict
 1. A class (classification)
 2. A quantity (a continious variable)
 3. An anomaly score
3. Your device must implement one or more ML/AI algorithms
 1. Depending on the nature and amount of your data, you may use either simple regression, tree or other Machine Learning Algorithms, that can be run on the Photon2, or a Neural Network based algorithm
 2. If you are working with time series data, preprocessing or filtering of you data may be nessecery
4. Your dataset
 1. You are to collect your own data from one or more sensors connected to the Photon2
 2. The data shall be labelled, and the lable stored in a suited manned with the data file
 3. The data should contain relevant metadata, such as samplerate, resolution, format
 4. Where possible, store the data in a standardised format such as *csv, wav, bmp, etc*

Non-functional requirements

1. Your data, software and optional hardware design must be shared
 1. You must create a public **github** account, and add relevant project files here.
 2. Hardware documentation, schematics, datasheets and optional pcb layouts are to be uploaded in **pdf** format.
 3. Software files are to be uploaded in raw source code format, e.g. **.C, CPP, h, .py**, etc.
 4. Data should be in suited format, e.g csv, wav, bmp, or similar.
2. The embedded platform shall be a Particle Photon2 device
 1. The platform shall have available digital or analog I/O to connect one or more sensors.
 2. Use the Photon2's Serial port or web-variables to communicate the current state of the observations.
 3. If possible, investigate optimisation by using the ARM CMSIS librarly for the Cortex M33 CPU in the Photon 2

Timeframe:

week 10 - approval of your project description and initial design requirements

week 12-15-18 Short presentations of the project progress

~week 19 - upload of your report, final source code and documentation to github etc.

At the exam:

The exam is oral

At the exam you are expected to give a short (maximum 5 minuttet) demo of your project, followed by a technical presentation of the project.

Since we have been reading the report, you should focus on presenting the key findings, or selected interesting topics at the exam, perhaps a novel way of solving a technical problem.

You can expect the project report to weigh app. 50% of your exam grade, and question in the remaining curriculum to weigh the remaining 50%